GDP per capita is a poor predictor of national well-being

Adrien Fabre (CNRS, CIRED)

January 2024

Introduction

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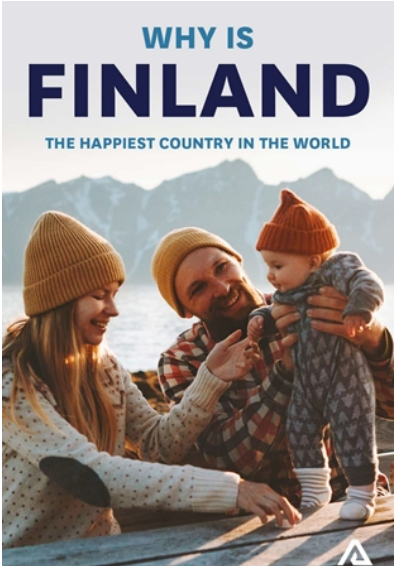

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Is money really buying happiness?

Literature

World Values Survey, $R^2 = .49$ (Inglehart & Klingemann, 2000)

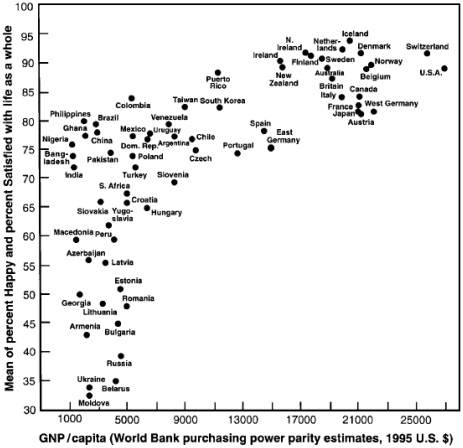

Figure 7.2
Subjective well-being by level of economic development ($R = 0.70$, $N = 65$, $p < 0.0000$). Source: World Values Surveys; GNP/capita purchasing power estimates from World Bank, World Development Report, 1997.

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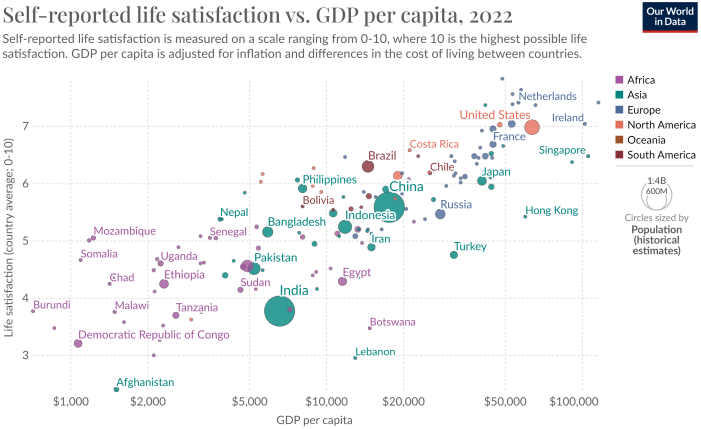
World Happiness Report (Gallup, 2023)



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We study new indicators and
challenge the view that national income is the best predictor of well-being.

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Another simple variable, the country's (macro) region, is a better predictor of national well-being.

Design

Data

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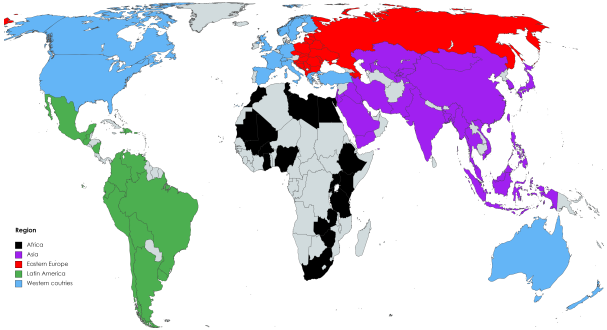

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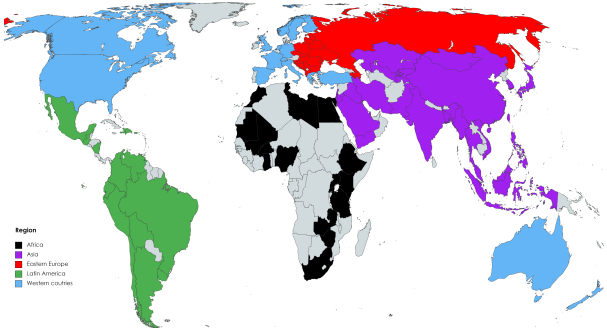

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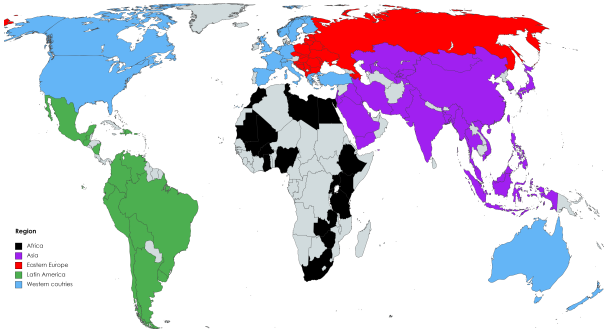

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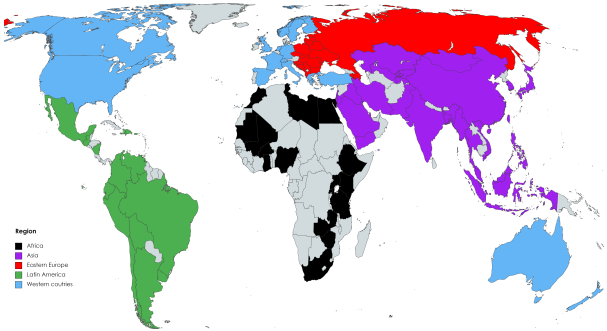
Very happy; *Quite happy*; *Not very happy*; *Not at all happy*; PNR



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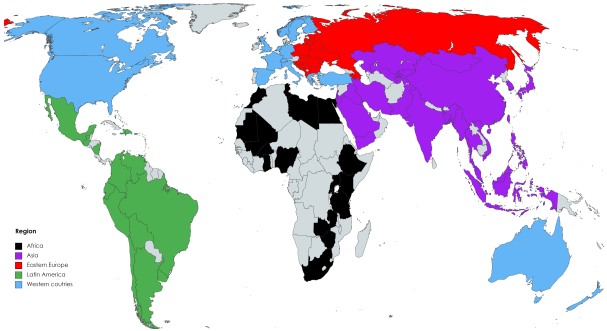
1-Completely dissatisfied – *10-Completeley satisfied*; PNR



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Bond & Lang (19) show that no single indicator can reliably identify two group's relative well-being, justifying reliance on several indicators.

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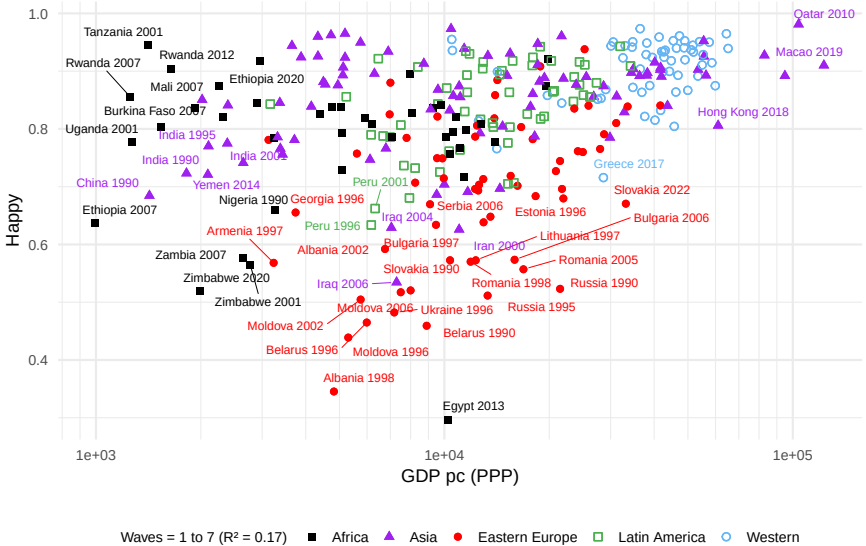
National well-being and income

Graphical evidence

Happy vs. log GDP p.c. (PPP) — All waves of WVS.



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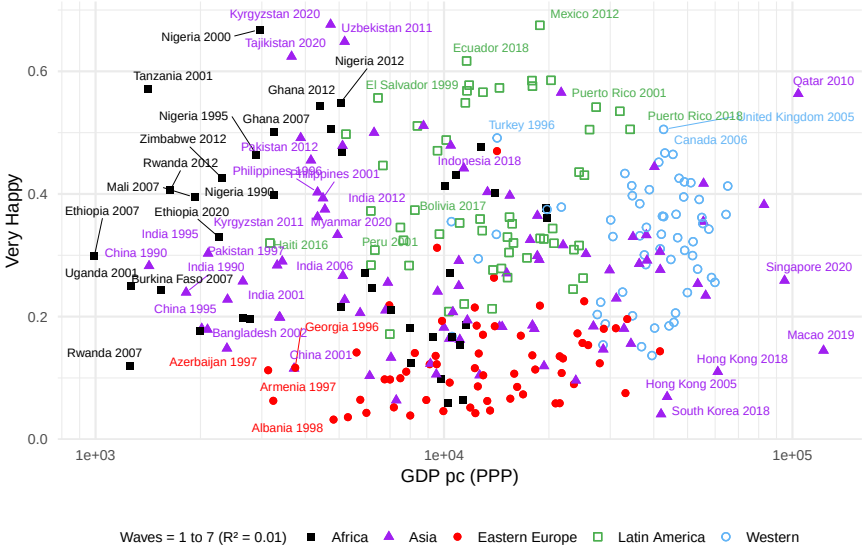
Very Happy vs. log GDP p.c. (PPP) — All waves of WVS.



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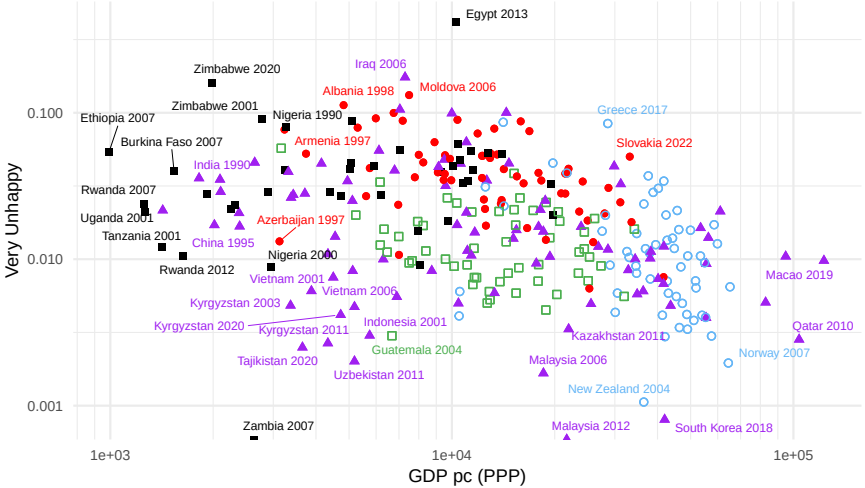
Very Unhappy vs. log GDP p.c. (PPP) — All waves of WVS.



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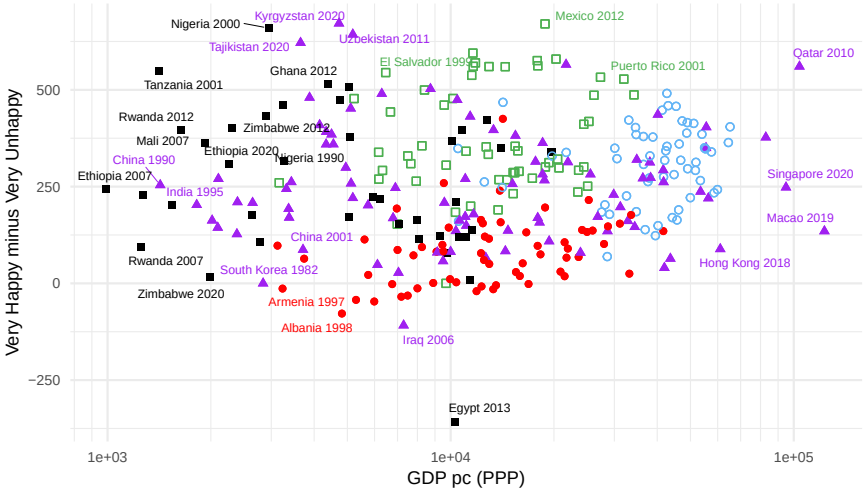

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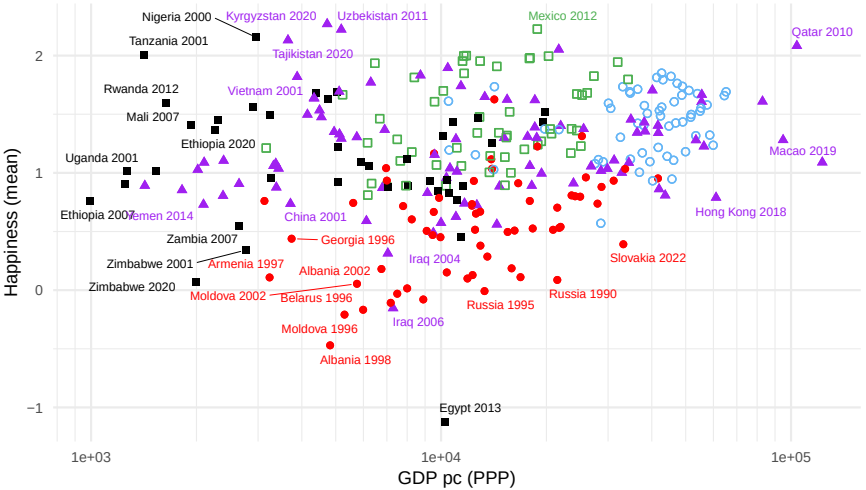
Happiness (mean) vs. log GDP p.c. (PPP) — All waves of WVS.



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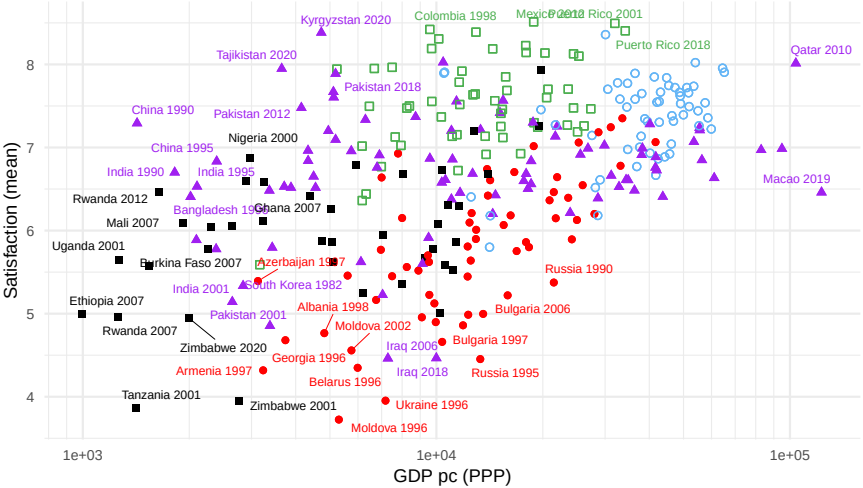
Satisfaction (mean) vs. log GDP p.c. (PPP) — All waves of WVS.



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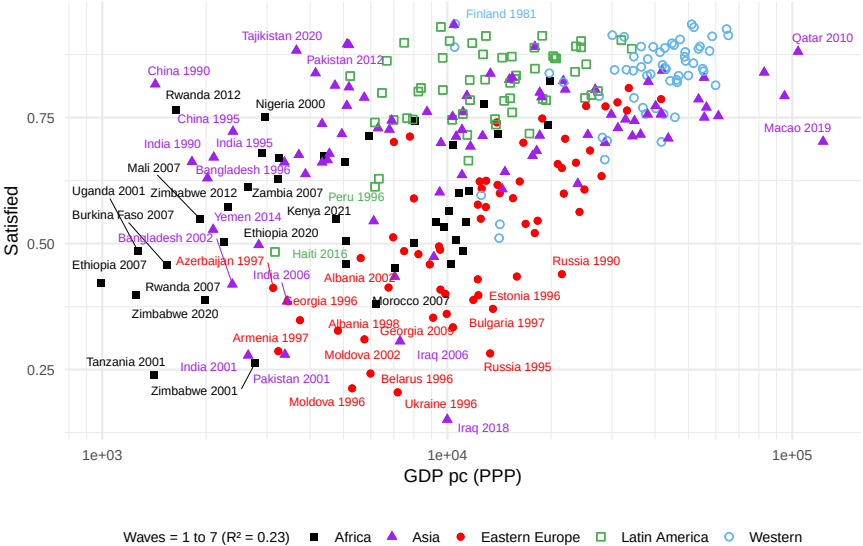

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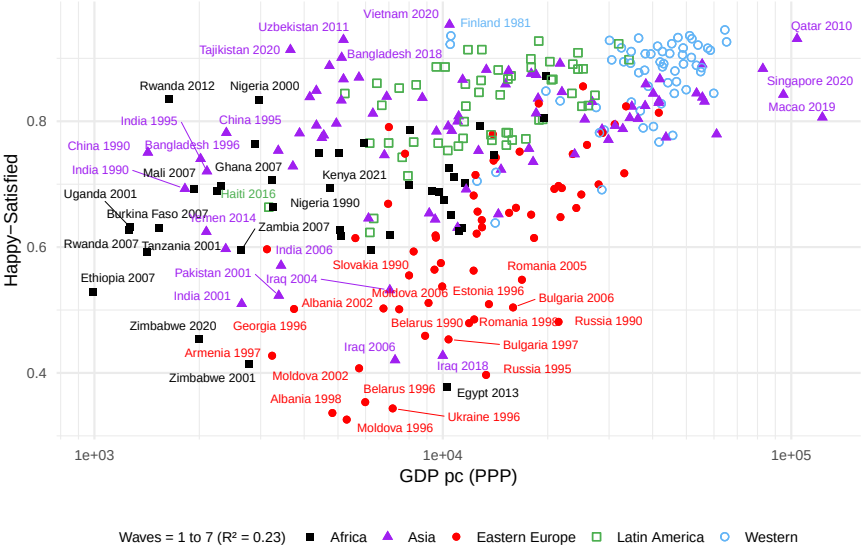
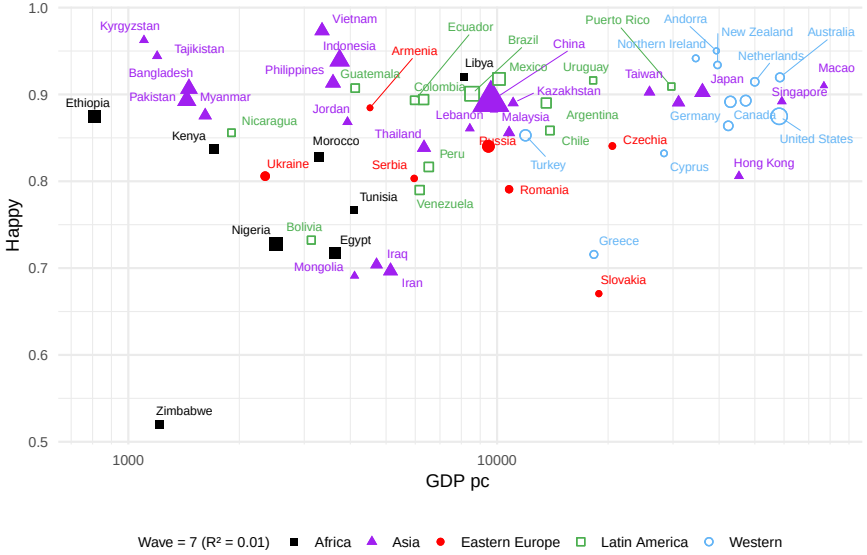
Happy + Satisfied vs. log GDP p.c. (PPP) — All waves of WVS.



Graphical evidence

Happy vs. log GDP p.c. (nominal) — Wave 7 (2017-22) of WVS, weighted by population.



Variance explained by GDP p.c.

More results

For different *well-being* and *income* indicators, we compute the R^2 of the regression:

$$\textit{well-being}_i = \alpha + \beta\, \textit{income}_i + u_i$$

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Happiness variable	log GDP p.c.		Income cluster						
	PPP	nominal	sextile PPP	k = 5 PPP	k = 6 PPP	k = 7 PPP	k = 7 nominal	Mean	Max
Satisfied	0.41	0.46	0.36	0.41	0.41	0.45	0.47	0.42	0.47
NA	0.24	0.32	0.24	0.27	0.27	0.28	0.32	0.28	0.32
NA	0.06	0.11	0.1	0.08	0.08	0.09	0.13	0.09	0.13
NA	0.01	0	0.06	0.03	0.03	0.04	0.02	0.03	0.06
NA	0.35	0.36	0.26	0.34	0.33	0.39	0.41	0.35	0.41
NA	0.33	0.32	0.26	0.34	0.31	0.38	0.38	0.33	0.38
Satisfaction (mean)	0.01	0.02	0.05	0.05	0.06	0.06	0.08	0.05	0.08
Mean	0.2	0.23	0.19	0.22	0.21	0.24	0.26	0.22	0.26
Max	0.41	0.46	0.36	0.41	0.41	0.45	0.47	0.42	0.47
Number of obs.	142	142	142	142	142	142	142		

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Happiness (mean) is poorly explained by income (8% at best).

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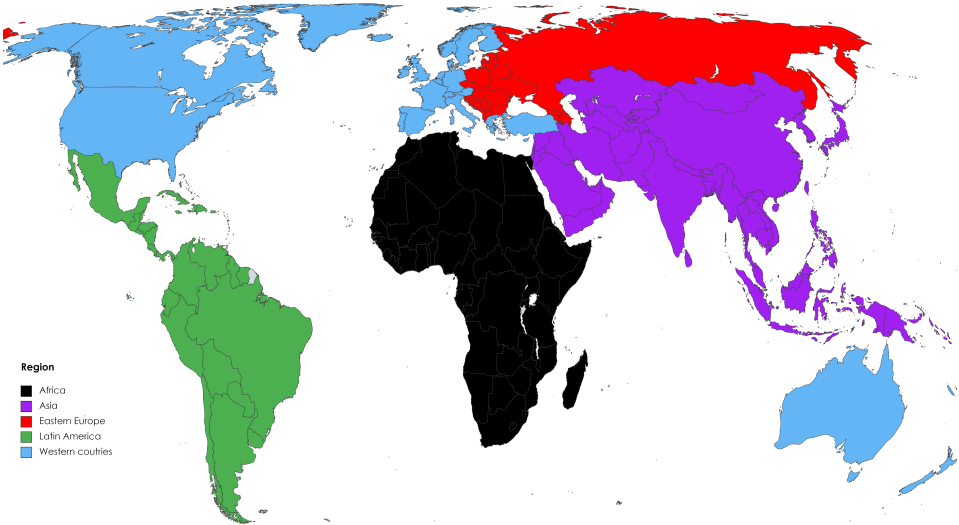
Blanchflower & Bryson (2023) show that on respective positive/negative affects, the happiest state is: Bhutan (well-rested), Denmark (satisfaction), Finland (anger), Hawaï (enjoy), Paraguay (smile), Taiwan (sadness), Uzbekistan (worry), Vietnam (pain).

Region vs. GDP per capita as predictor of well-being

Region grouping

WVS countries grouped into the five UN regional groups.



Comparing the share of variance explained by income vs. region

For different *well-being* and *income* indicators, we run regressions and compute corresponding R^2:

$$\textit{well-being}_i = \alpha_1 + \beta_1 \, \textit{income}_i + u_i \tag{1}$$

$$\textit{well-being}_i = \alpha_2 + \gamma_2 \, \textit{region}_i + e_i \tag{2}$$

$$\textit{well-being}_i = \alpha_3 + \beta_3 \, \textit{income}_i + \gamma_3 \, \textit{region}_i + \varepsilon_i \tag{3}$$

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Comparing the share of variance explained by income vs. region

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This follows the LMG methodoly (Lindeman, Merenda & Gold, 1980; Grömping, 2007).

Share of explained variance that is explained by income

More results

Happiness variable	log GDP p.c.		Income cluster						
	PPP	nominal	sextile PPP	k = 5 PPP	k = 6 PPP	k = 7 PPP	k = 7 nominal	Mean	Max
Satisfied	0.35	0.42	0.35	0.36	0.37	0.35	0.42	0.38	0.42
NA	0.21	0.28	0.2	0.22	0.23	0.21	0.28	0.23	0.28
NA	0.04	0.06	0.04	0.06	0.08	0.06	0.08	0.06	0.08
NA	0	0	0.07	0.07	0.09	0.07	0.06	0.05	0.09
NA	0.33	0.38	0.34	0.34	0.35	0.33	0.4	0.35	0.4
NA	0.39	0.42	0.39	0.41	0.41	0.39	0.46	0.41	0.46
Satisfaction (mean)	0.26	0.31	0.24	0.26	0.27	0.25	0.31	0.27	0.31
Mean	0.22	0.27	0.23	0.25	0.26	0.24	0.29	0.25	0.29
Max	0.39	0.42	0.39	0.41	0.41	0.39	0.46	0.41	0.46
Number of obs.	304	304	304	304	304	304	304		

Region is a better predictor of national well-being than income

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This indicator explains 19% of the explained variance for **Happiness** and 32% for **Satisfaction**.

Region is a better predictor than region in 94% of alternative specifications: looking at each wave separately, weighting countries by population, dropping pandemic years...
(including 86% of 88 specifications involving the best-predicting income variable). ▸ More results

Conclusion

Take away and future research

National well-being is more correlated with the world region than with the GDP p.c.

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Despite evidence against translation issues (Diener & Suh, 2000),
We should check whether emotions are better predicted by region than income.

Robustness checks

Variance explained by PPP income cluster (k = 7)

Go back

Happiness variable	All waves	Only selected waves							
	Pop. weight	1 & 2	3	4	5	6	7	Mean	Max
Very Happy	0.05	0.25	0.06	0.17	0.06	0.12	0.21	0.13	0.25
Happy	0.21	0.19	0.24	0.23	0.22	0.17	0.06	0.19	0.24
Very Unhappy	0.04	0.2	0.15	0.19	0.16	0.1	0.08	0.13	0.2
Satisfied	0.23	0.2	0.22	0.35	0.26	0.23	0.1	0.23	0.35
Satisfaction (mean)	0.16	0.23	0.17	0.32	0.2	0.21	0.05	0.19	0.32
Happiness (mean)	0.09	0.18	0.13	0.22	0.15	0.14	0.07	0.14	0.22
Happy + Satisfied	0.27	0.2	0.25	0.33	0.27	0.21	0.09	0.23	0.33
V. Happy – V. Unhappy	0.05	0.16	0.07	0.19	0.08	0.12	0.16	0.12	0.19
Mean	0.14	0.2	0.16	0.25	0.18	0.16	0.1	0.17	0.25
Max	0.27	0.25	0.25	0.35	0.27	0.23	0.21	0.23	0.35
Number of obs.	304	26	56	40	58	60	64		

Share of explained variance that is explained by PPP income cluster (k = 7)

Go back

Happiness variable	All waves	Only selected waves							
	Pop. weight	1 & 2	3	4	5	6	7	Mean	Max
Very Happy	0.19	0.3	0.08	0.36	0.13	0.37	0.47	0.27	0.47
Happy	0.54	0.33	0.36	0.58	0.39	0.48	0.26	0.42	0.58
Very Unhappy	0.25	0.26	0.28	0.57	0.44	0.43	0.34	0.37	0.57
Satisfied	0.57	0.35	0.28	0.56	0.38	0.42	0.25	0.4	0.57
Satisfaction (mean)	0.36	0.37	0.22	0.47	0.3	0.38	0.12	0.32	0.47
Happiness (mean)	0.31	0.25	0.18	0.46	0.25	0.43	0.23	0.3	0.46
Happy + Satisfied	0.57	0.32	0.32	0.57	0.39	0.42	0.24	0.41	0.57
V. Happy – V. Unhappy	0.22	0.22	0.1	0.38	0.16	0.41	0.38	0.27	0.41
Mean	0.38	0.3	0.23	0.5	0.3	0.42	0.29	0.34	0.5
Max	0.57	0.37	0.36	0.58	0.44	0.48	0.47	0.42	0.58
Number of obs.	304	26	56	40	58	60	64		